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Surgical and Radiographic Anatomy of the Adrenals

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Keywords

adrenal gland; radiographic anatomy; surgical anatomy

Questions

1. Which of the following statements is (are) TRUE?
    a. The weight of each of the glands is approximately 10 g.
    b. The adrenal glands are in close proximity to the crus of the diaphragm.
    c. The right gland is crescentic in shape.
    d. The left adrenal gland lies adjacent to the splenic vessels.
    e. Both b and d.
2. Which of the following statements is (are) TRUE?
    a. The right renal vein is longer than the left renal vein.
    b. The right adrenal vein is longer than the left adrenal vein.
    c. The right kidney is typically located lower in the retroperitoneum than the left kidney.
    d. The right adrenal gland is typically located higher in the retroperitoneum than the left adrenal gland.
    e. Both c and d.
3. The adrenal arteries are branches from:
    a. the aorta.

b. the inferior phrenic arteries.
c. the renal arteries.
d. the celiac arterial trunk.
e. a, b, and c.

4. In cases of renal ectopia, the ipsilateral adrenal gland is typically:
   a. absent.
   b. found in its normal anatomic position in the upper retroperitoneum.
   c. found in association with the contralateral adrenal gland.
   d. found closely applied to the superior pole of the ectopic kidney.
   e. found closely associated with the ipsilateral renal artery.
5. In cases of unilateral renal agenesis, the ipsilateral adrenal gland is commonly:
   a. absent.
   b. found in its normal anatomic position in the upper retroperitoneum.
   c. found in association with the contralateral adrenal gland.
   d. found just inside the ipsilateral internal inguinal ring.
   e. found in an ectopic, intrathoracic location.
6. As one proceeds outward from the adrenal medulla, the three separate functional layers of the adrenal cortex, in correct order, are:
   a. zona reticularis, zona fasciculata, and zona glomerulosa.
   b. zona fasciculata, zona reticularis, and zona glomerulosa.
   c. zona glomerulosa, zona fasciculata, and zona reticularis.
   d. zona glomerulosa, zona reticularis, and zona fasciculata.
   e. zona reticularis, zona glomerulosa, and zona fasciculata.
7. Which of the following statements is (are) FALSE?
   a. The adrenal medulla produces catecholamines in response to stimulation from the sympathetic nervous system.
   b. The zona glomerulosa produces aldosterone in response to angiotensin II.
   c. The zona fasciculata of the adrenal cortex produces glucocorticoids in response to adrenocorticotropic hormone (ACTH).

d. The zona reticularis of the adrenal cortex produces androgens in response to luteinizing hormone (LH).
e. Both b and c.

8. Which of the following statements is TRUE?
    a. Ultrasound imaging cannot differentiate between solid and cystic masses of the adrenal gland.
    b. Contrast resolution of magnetic resonance imaging (MRI) is inferior to that of computed tomography (CT) in enabling differentiation of adrenal masses.
    c. CT is the most widely used modality for imaging the adrenal glands.
    d. Normal adrenal tissue has a density of greater than 10 Hounsfield units on noncontrast CT imaging.
    e. MRI, via T1-weighted and T2-weighted images, can provide functional data about adrenal masses.

Answers

1. **e. Both b and d. Both adrenal glands are in close proximity to the crus of the diaphragm.** The left adrenal gland is in contact with the medial aspect of the upper pole of the left kidney. The adjacent structures include the splenic vessels and body of the pancreas anteriorly, the aorta medially, and the psoas muscle posteriorly.
2. **e. Both c and d.** The right kidney is typically located lower in the retroperitoneum than the left kidney. **The right adrenal gland lies more superiorly in the retroperitoneum than the left adrenal gland.**
3. **e. a, b, and c.** The three main adrenal arteries each branch into cascades of 10 to 50 smaller arteries, which then supply each adrenal gland. The three major arterial sources for each gland are: (1) superior branches from the inferior phrenic artery, (2) middle branches directly from the aorta, and (3) inferior branches from the ipsilateral renal artery.

4. **b. Found in its normal anatomic position in the upper retroperitoneum. In cases of renal ectopia, the adrenal gland is usually found in approximately its normal anatomic position.**
5. **b. Found in its normal anatomic position in the upper retroperitoneum.** In cases of renal agenesis, the adrenal gland on the involved side is typically present.
6. **a. Zona reticularis, zona fasciculata, and zona glomerulosa.** The adult adrenal gland consists of an outer cortex, which makes up approximately 90% of the adrenal mass, and an inner medulla. The adrenal cortex consists of three concentric zones: the outer zona glomerulosa comprising approximately 15% of the cortex, the middle zona fasciculata 80% of the cortex, and the inner zona reticularis 5% to 7% of the cortex. **The zona glomerulosa produces aldosterone, zona fasciculata produces glucocorticoids, and zona reticularis produces sex steroids.**
7. **d. The zona reticularis of the adrenal cortex produces androgens in response to luteinizing hormone (LH). The production of androgen by the zona reticularis is regulated by pituitary release of ACTH.**
8. **c. CT is the most widely used modality for imaging the adrenal glands.** Ultrasound imaging can be used to differentiate between solid and cystic adrenal masses. **Normal adrenal tissue has a density of less than or equal to 10 Hounsfield units (HU) on noncontrast CT imaging.** MRI, via T1-weighted and T2-weighted images, is superior to CT and enables differentiation of adrenal masses. Although CT and MRI provide excellent morphologic information, these studies cannot provide functional data.

Chapter Review

1. The adrenals are enclosed within the Gerota fascia.
2. The right adrenal vein is short and enters the vena cava posterolaterally.
3. Adrenal rests are found in proximity to the celiac axis and along the path of gonadal descent. In patients with congenital adrenal

hyperplasia, adrenal rests along the spermatic cord may become quite large.

4. The right kidney is typically located lower in the retroperitoneum than the left kidney. The right adrenal gland lies more superiorly in the retroperitoneum than the left adrenal gland. Both adrenal glands are in close proximity to the crus of the diaphragm.
5. In cases of renal ectopia, the adrenal gland is usually found in approximately its normal anatomic position.
6. The zona glomerulosa produces aldosterone, the zona fasciculata produces glucocorticoids, and the zona reticularis produces sex steroids.
7. The production of androgen by the zona reticularis is regulated by pituitary release of adrenocorticotropic hormone (ACTH).
8. Normal adrenal tissue has a density of less than or equal to 10 Hounsfield units (HU) on noncontrast CT imaging.

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Pathophysiology, Evaluation, and Medical Management of Adrenal Disorders

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Keywords

adrenal; adrenalectomy; adrenocortical carcinoma; Conn; Cushing; gland; mass; pheochromocytoma

Questions

1. At birth the adrenal cortex:
    a. is completely developed.
    b. weighs half as much as the adrenal cortex in adults.
    c. is composed of fetal and adult components.
    d. will continue to enlarge until 12 months of age.
    e. is composed of a single histologic zone.
2. Adrenal rest tissue within the testis can mimic testicular cancer in patients with:
    a. neuroblastoma.
    b. pheochromocytoma.
    c. primary aldosteronism.
    d. congenital adrenal hyperplasia.
    e. cryptorchidism.
3. In cases of renal agenesis, the ipsilateral adrenal gland is typically:

a. absent.
b. in the normal location.
c. located at the level of the eighth thoracic vertebral body.
d. located at the level of the first lumbar vertebral body.
e. at a location dependent on the cause of renal agenesis.

4. The most abundant product of the adrenal cortex is:
a. mineralocorticoids.
b. glucocorticoids.
c. adrenal androgens.
d. catecholamines.
e. adrenocorticotropic hormone (ACTH).

5. Aldosterone synthase (CYP11B2) is unique to:
a. the zona glomerulosa.
b. the zona fasciculata.
c. the zona reticularis.
d. the adrenal medulla.
e. the distal renal tubule.

6. The only zone of the adrenal cortex that does not atrophy upon pituitary failure is:
a. the zona glomerulosa.
b. the zona fasciculata.
c. the zona reticularis.
d. the adrenal medulla.
e. none of the above.

7. Presence of the phenylethanolamine-*N*-methyltransferase (PNMT) enzyme in the adrenal medulla is significant because:
a. the enzyme catalyzes degradation of catecholamines to metanephrines.
b. the enzyme catalyzes conversion of catecholamines to vanillylmandelic acid (VMA).
c. the enzyme converts tyrosine to dopamine.
d. the enzyme catalyzes the conversion of norepinephrine to epinephrine.
e. all of the above.

8. Metanephrines:

a. refers to the term used for catecholamines and their byproducts.
b. refers to the combined term for methylated metabolites of norepinephrine (normetanephrine) and epinephrine (metanephrine).
c. refers to precursors to normetanephrines.
d. are rarely helpful in establishing a diagnosis of pheochromocytoma.
e. refers to the term used to describe epinephrine and norepinephrine in the context of pheochromocytoma symptomatology.

9. The term *free metanephrines:*
a. is interchangeable with the term *total metanephrines*.
b. is interchangeable with the term *fractionated metanephrines.*
c. refers to normetanephrine and metanephrine that are not conjugated by a sulfate moiety.
d. refers to normetanephrine and metanephrine that are not bound to albumin.
e. refers to all of the above.

10. The most common cause of Cushing syndrome (exclusive of exogenous steroid intake) is:
a. Cushing disease.
b. a cortisol-producing adrenal adenoma.
c. ectopic ACTH production by a lung malignancy.
d. an adrenal carcinoma.
e. a pheochromocytoma.

11. What common urologic ailment can be found in up to 50% of patients with Cushing syndrome?
a. Testicular cancer
b. Torsion of the appendix testis
c. Urolithiasis
d. Fournier gangrene
e. Stress urinary incontinence

12. How does one perform a low-dose dexamethasone suppression test (LDDST)?

a. Admit the patient and measure serum cortisol levels every 6 hours while the patient is on a dexamethasone drip
b. Measure the patient's saliva cortisol level at midnight
c. Obtain a 24-hour urine cortisol measurement after the patient receives 1 mg of dexamethasone with the first void
d. Have the patient take 10 mg of dexamethasone at 11 PM and measure urinary cortisol the next morning
e. Have the patient take 1 mg of dexamethasone at 11 PM and measure serum cortisol the next morning

13. The adrenal surgeon plays no role in the management of Cushing disease.
    a. True
    b. False
14. What percentage of patients presenting with primary aldosteronism are hypokalemic?
    a. 5% to 12%
    b. 9% to 37%
    c. 33% to 50%
    d. 55% to 75%
    e. 63% to 91%
15. Elevated aldosterone in patients with familial hyperaldosteronism type I is mediated by:
    a. renin.
    b. sodium.
    c. angiotensin II.
    d. cortisol.
    e. ACTH.
16. What percentage of patients with a positive screening test will be diagnosed with primary aldosteronism after confirmatory testing?
    a. 2% to 10%
    b. 20% to 40%
    c. 30% to 50%
    d. 50% to 70%
    e. 75% to 90%
17. What is the most common subtype of primary aldosteronism?
    a. Idiopathic hyperplasia

b. Aldosterone-producing adenoma
c. Unilateral adrenal hyperplasia
d. Familial hyperaldosteronism type I
e. Adrenocortical carcinoma

18. The primary determinant of potentially surgically correctable primary aldosteronism is:
    a. blood pressure.
    b. patient age.
    c. demonstration of lateralized aldosterone secretion.
    d. response to medical therapy.
    e. plasma aldosterone levels.
19. Which class of antihypertensives is contraindicated during the evaluation of primary aldosteronism?
    a. Calcium channel blockers
    b. Alpha blockers
    c. Beta blockers
    d. Aldosterone-receptor blockers
    e. Angiotensin-converting enzyme inhibitors
20. What percentage of patients with incidental adrenal masses prove to have pheochromocytoma?
    a. 1%
    b. 5%
    c. 10%
    d. 25%
    e. 35%
21. What subtype is assigned to patients with a von Hippel-Lindau *(VHL)* mutation and a history of pheochromocytoma but no other stigmata of the VHL syndrome?
    a. Type 1
    b. Type 2A
    c. Type 2B
    d. Type 2C
    e. None of the above
22. What genetic abnormality is strongly linked with malignant pheochromocytoma?
    a. *RET* mutation

b. *VHL* mutation
c. *SDHB* mutation
d. *SDHD* mutation
e. All of the above

23. What test is considered the cornerstone for modern pheochromocytoma biochemical testing?
    a. Plasma catecholamines
    b. Plasma-free metanephrines or fractionated urinary metanephrines
    c. VMA testing
    d. Fasting morning urinary norepinephrine
    e. Adrenal vein sampling for catecholamines
24. With regard to preoperative pheochromocytoma blockade, β-blockers should be started:
    a. 2 weeks prior to adrenalectomy.
    b. at least several days prior to α-blockers.
    c. to control tachycardia and arrhythmias that can result upon initiation of α blockade.
    d. in conjunction with metyrosine.
    e. never, because β-blockers can be lethal in patients with pheochromocytomas.
25. Patients with adrenal crisis can exhibit all of the following symptoms EXCEPT:
    a. hypotension unresponsive to fluid resuscitation.
    b. abdominal pain.
    c. nausea.
    d. fever.
    e. priapism.
26. All of the following lesions can be extra-adrenal EXCEPT:
    a. myelolipoma.
    b. ganglioneuroma.
    c. aldosteronoma.
    d. pheochromocytoma.
    e. oncocytoma.
27. A 25-year-old woman is diagnosed with a left adrenal mass, with abundant stippled calcifications, that exhibits imaging features

inconsistent with adrenal adenoma. The patient complains of a severe bout of diarrhea that started approximately 8 months ago. Adrenalectomy reveals a ganglioneuroma. At her postoperative visit she is grateful, because her gastrointestinal (GI) complaints vanished following surgery. The substance responsible for diarrhea in this patient is most likely:

a. metanephrine.
b. epinephrine.
c. norepinephrine.
d. VMA.
e. vasoactive intestinal polypeptide (VIP).

28. What percentage of adrenal cysts is associated with malignancy in surgical series?

a. 1%
b. 3%
c. 7%
d. 12%
e. 15%

29. Adrenocortical carcinoma in children:

a. has a more favorable 5-year survival rate compared with adults.
b. uses the same pathologic staging system as adults.
c. is rarely associated with virilization.
d. has equal female and male incidence in children older than 10 years.
e. is frequently metastatic to the central nervous system.

30. The most common hormone secreted by adrenocortical carcinoma is:

a. aldosterone.
b. testosterone.
c. dehydroepiandrosterone (DHEA).
d. cortisol.
e. androstenedione.

31. The Weiss criteria for identifying malignant adrenal tumors should be applied with caution in tumors with:

a. necrosis.

b. high mitotic index.
c. inferior vena cava (IVC) invasion.
d. liver metastasis.
e. oncocytic features.

32. In patients presenting with metastatic adrenocortical carcinoma, the systemic agent of choice is:
    a. valrubicin.
    b. mitotane alone or in combination with additional cytotoxic agents.
    c. gemcitabine and cisplatin.
    d. docetaxel (Taxotere).
    e. bleomycin, etoposide, and cisplatin.
33. In the treatment of pathologically localized adrenocortical carcinoma:
    a. adjuvant radiation therapy decreases systemic progression.
    b. complete surgical resection offers the best chance of cure.
    c. increased Ki-67 expression has been associated with improved survival.
    d. the tumor's functional status is an independent predictor of survival.
    e. adjuvant therapy with mitotane has no proven benefit.
34. A 50% false-positive rate can be seen during low-dose dexamethasone suppression testing in:
    a. men with testicular cancer.
    b. women taking oral contraceptives.
    c. men with history of orchiopexy.
    d. patients with brain malignancy.
    e. patients with pheochromocytoma.
35. The adrenal gland should be resected whenever one performs a radical nephrectomy.
    a. True
    b. False
36. A 55-year-old woman presents for the evaluation of an adrenal mass. Past medical history is significant for severe hypertension requiring four oral medications for adequate blood pressure control. A noncontrast computed tomography (CT) scan of the abdomen

reveals a 3-cm left adrenal mass, with an average attenuation of 7 Hounsfield units (HU). Appropriate initial screening should include:

a. free-fractionated plasma metanephrines, plasma aldosterone concentration, and plasma renin activity.
b. LDDST and serum catecholamines.
c. late-night salivary cortisol test, plasma aldosterone concentration, plasma renin activity, and plasma free metanephrines.
d. plasma aldosterone concentration and plasma renin activity.
e. 24-hour urinary-fractionated metanephrines and serum cortisol concentration.

37. A 62-year-old man presents for postoperative surveillance of renal cell carcinoma. Three years prior, a right nephrectomy was performed for T2N0M0 grade III clear cell renal cell carcinoma. A current abdominal CT scan reveals a 3.5-cm right adrenal mass with an average attenuation of 32 HU prior to contrast administration. A CT washout study is performed, and absolute percent washout is calculated to be 52%. No other suspicious lesions are noted within the chest, abdomen, or pelvis. The next step in management should be:

a. observation.
b. adrenalectomy.
c. percutaneous biopsy.
d. initiation of an oral tyrosine kinase inhibitor.
e. assessment of the adrenal tumor's functional status.

38. A 58-year-old woman has been diagnosed with primary aldosteronism based on appropriate screening and confirmatory testing. A CT scan of the abdomen reveals a 1.0-cm left adrenal mass. Adrenal vein sampling is performed with the results outlined below. Results of adrenal vein sampling:

Right cortisol gradient 2.7:1
Left cortisol gradient 3.4:1
Aldosterone ratio (left:right) 2.5:1
The next step in management would be:

a. repeat adrenal vein sampling with ACTH stimulation.

b. counseling for left adrenalectomy.
c. counseling for right adrenalectomy.
d. initiation of medical management based on diagnosis of bilateral adrenal hyperplasia.
e. ^{131}I-iodomethyl-norcholesterol (NP-59) scintigraphy to confirm lateralization.

39. A 43-year-old woman undergoes an uneventful right laparoscopic adrenalectomy for a 4.5-cm pheochromocytoma. The pathology report states that the lesion is benign and that the margins are negative. On postoperative day 1, she is ready for discharge and wants to know if any additional follow-up is necessary. You inform her that the following is required:
    a. Consideration of genetic screening
    b. Repeat metabolic testing in 2 weeks
    c. Cross-sectional imaging in 6 months
    d. Biochemical testing in 6 months and then lifelong biochemical testing
    e. All of the above

Pathology

1. A 60-year-old woman is noted to have a 4-cm left adrenal mass on an abdominal CT scan. The endocrine workup is negative, and the mass is excised laparoscopically. The pathology, depicted in Fig. 106.1, is reported as a myolipoma. The next step in management is:

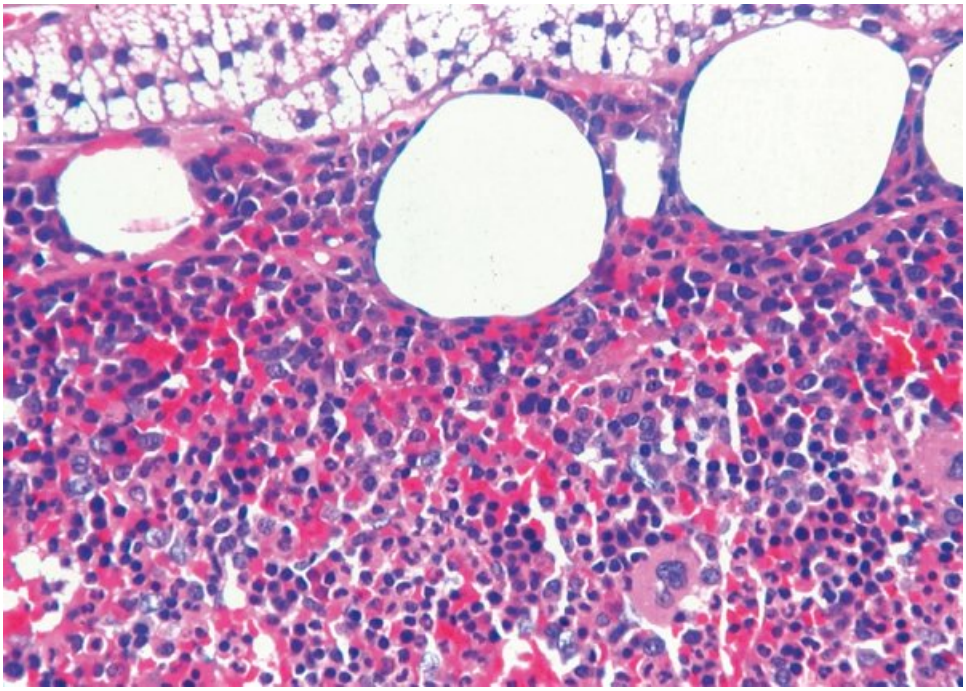


FIG. 106.1

From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Edinburgh, 2008, Mosby.

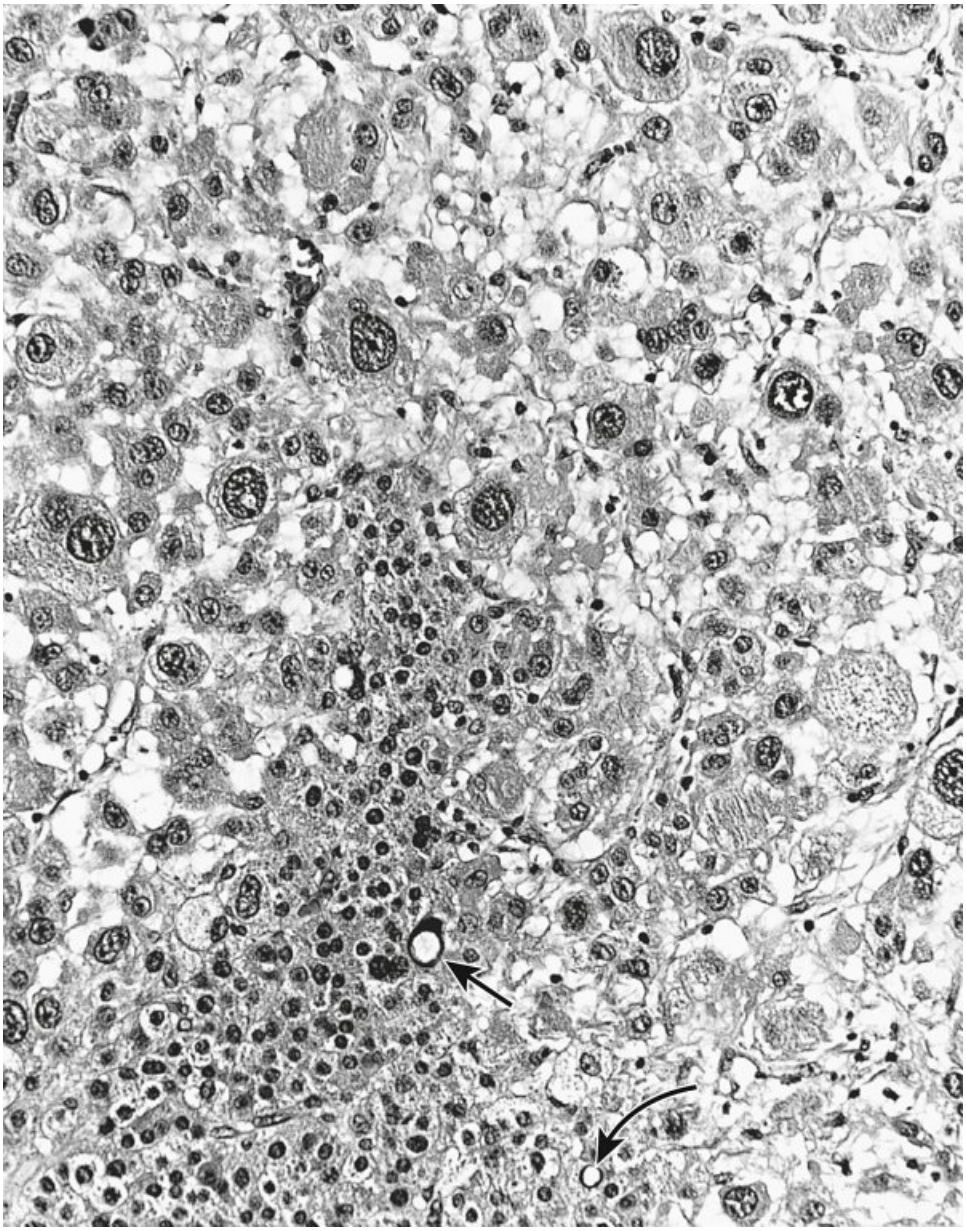


FIG. 106.2
From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Edinburgh, 2008, Mosby.

a. no additional therapy is indicated because this is a benign tumor.
b. ask the pathologist to grade the tumor.
c. the patient should receive mitotane.
d. the patient should be followed carefully for development of hypertension.

e. a metaiodobenzylguanidine (MIBG) scan should be obtained.

2. A 45-year-old man has an incidentally discovered 6-cm adrenal mass. Hormonal workup is negative. The mass is laparoscopically removed. The pathology is depicted in Fig. 106.2 and is read as adrenal corticocarcinoma. On separate stains the lesion stains positive for Ki-67. The next step in management is:
   a. request a pathologic grade.
   b. stain for catecholamines.
   c. inquire as to whether any fat was observed in the specimen.

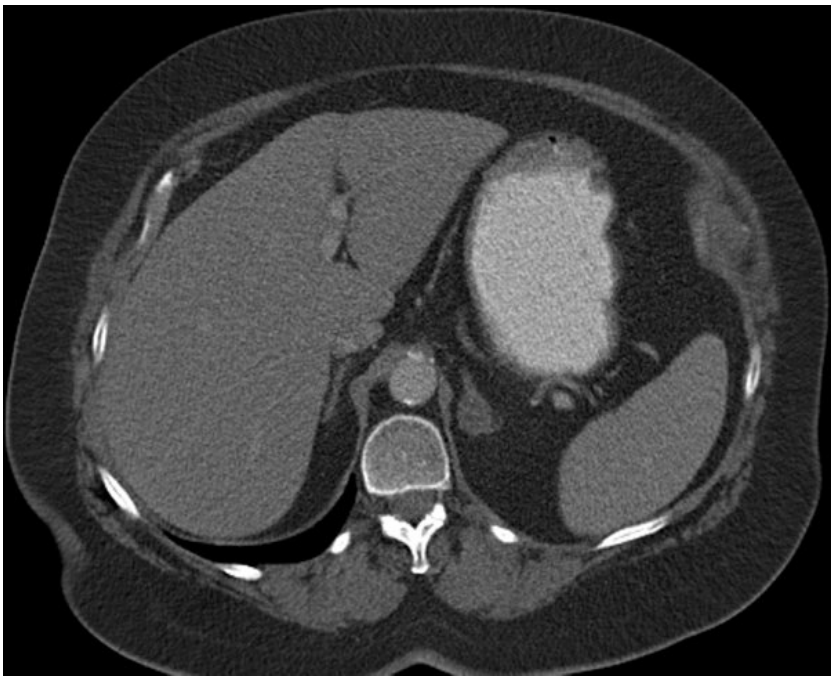


FIG. 106.3

   d. short-term follow-up with imaging.
   e. adjuvant mitotane.

Imaging

1. See Fig. 106.3. A 42-year-old man with lung cancer has this CT scan. The right adrenal nodule has attenuation measurements of 7 HU. The most likely diagnosis is:

a. adrenal adenoma.
b. adrenal metastasis.
c. indeterminate adrenal nodule.
d. pheochromocytoma.
e. adrenal myelolipoma.

Answers

1. **c. Is composed of fetal and adult components.** The adrenal gland weighs twice as much as the adult gland and begins to atrophy at birth. Development of the gland continues during the first 3 years of life, with the zona reticularis developing last.
2. **d. Congenital adrenal hyperplasia.** Testicular adrenal rests must be remembered when evaluating patients with congenital adrenal hyperplasia and testicular masses to avoid an unnecessary orchiectomy.
3. **b. In the normal location.** Reports of adrenal agenesis are extremely rare.
4. **c. Adrenal androgens.** Although adrenal androgens are arguably the least physiologically significant compounds produced by the adrenals, the glands produce more than 20 mg of these compounds per day. Meanwhile, only 100 to 150 mcg/d of aldosterone and approximately 10 to 20 mg/d of cortisol are produced by the glands.
5. **a. The zona glomerulosa. The zona glomerulosa cells are the sole source of aldosterone in humans.**
6. **a. The zona glomerulosa. Production of aldosterone in the zona glomerulosa is primarily regulated by angiotensin II through the renin-angiotensin-aldosterone system and potassium levels.** Elevation of ACTH can also increase aldosterone secretion, but this is a much less potent stimulus. Therefore, in pituitary failure, when ACTH levels fall, the zona glomerulosa fails to atrophy.
7. **d. The enzyme catalyzes the conversion of norepinephrine to epinephrine.** PNMT is virtually unique to the adrenal medulla (the brain and organ of Zuckerkandl also express the protein).

Therefore the presence of PNMT results in epi nephrine being virtually a unique product of the adrenal gland.

8. **b. Refers to the combined term for methylated metabolites of norepinephrine (normetanephrine) and epinephrine (metanephrine).** The enzyme catechol-*O*-methyltransferase catalyzes the methylation of norepinephrine to normetanephrine and epinephrine to metanephrine. The term *normetanephrines* is not used.
9. **c. Refers to normetanephrine and metanephrine that are not conjugated by a sulfate moiety.** Free metanephrines are unsulfonated normetanephrine and metanephrine, whereas total metanephrines refers to both conjugated and free compounds. The term *fractionated metanephrines* refers to laboratory reports that differentiate between metanephrine and normetanephrine concentrations (i.e., instead of only reporting metanephrine concentration, state the concentration of metanephrine and normetanephrine separately).
10. **a. Cushing disease. Cushing disease, which describes overproduction of ACTH by the pituitary, accounts for some 70% of endogenous Cushing syndrome.**
11. **c. Urolithiasis. Urolithiasis is seen in up to 50% of patients with Cushing syndrome; therefore stone formers with cushingoid features deserve a hypercortisolemia workup. The astute urologist should also remember that Cushing patients can also exhibit hypogonadal hypogonadism and should have a low threshold for a hypercortisolism workup in men with low testosterone and low gonadotropin levels.**
12. **e. Have the patient take 1 mg of dexamethasone at 11 pm and measure serum cortisol the next morning.** Despite its intimidating name and rather complex physiologic underpinnings, the test is remarkably simple to administer. Write the patient a prescription for 1 mg of dexamethasone and ask that it be taken by mouth at 11 PM. The next morning, determine the patient's serum cortisol level. If the cortisol level is 5 mcg/dL or higher (i.e., not suppressed), then the patient likely has hypercortisolemia. Be aware that women on birth control will have false-positive results.

13. **b. False.** ACTH-secreting pituitary adenomas are treated with transsphenoidal surgical resection. However, a cure is seen in only 60% to 80% of patients. Among those who are cured, there is approximately 25% relapse. **One option for patients who are refractory to neurosurgical treatment is a bilateral adrenalectomy.** However, this treatment should not be performed hastily. It is crucial that a thoughtful multidisciplinary decision be made. **Up to 30% of patients with Cushing disease who undergo bilateral adrenalectomy may develop Nelson syndrome—progressive growth of the pituitary adenoma causing increased intracranial pressure and compression of the ocular chiasm.**
14. **b. 9% to 37%**. **Although hypokalemia has been classically described as a common finding in primary aldosteronism, only 9% to 37% of newly diagnosed patients are hypokalemic.**
15. **e. ACTH.** Because of the chimeric fusion of the promoter region of 11β-hydroxylase and the coding region of aldosterone synthase, aldosterone production is mediated by ACTH in familial hyperaldosteronism type I.
16. **d. 50% to 70%.** Fifty percent to 70% of patients with a positive screening test will be diagnosed with primary aldosteronism following confirmatory testing.
17. **a. Idiopathic hyperplasia.** This subtype of primary aldosteronism accounts for approximately 60% of cases.
18. **c. Demonstration of lateralized aldosterone secretion.** Lateralization of aldosterone secretion is the primary determinant of successful surgical treatment of primary aldosteronism.
19. **d. Aldosterone-receptor blockers.** Aldosterone-receptor blockers (spironolactone and eplerenone) are contraindicated during the evaluation of primary aldosteronism. Patients requiring these agents for blood pressure control should be transitioned to other medications during testing for at least 6 weeks.
20. **b. 5%.** Incidentally discovered lesions account for 10% to 25% of all pheochromocytomas.
21. **d. Type 2C.** Type 1 = VHL patient with no evidence or family history of pheochromocytoma. Type 2 patients are those with evidence or family history of pheochromocytoma. Type 2 is further

subdivided. Type 2A = patients with concomitant RCC; type 2B = no evidence of renal malignancy; type 2C = patients with pheochromocytoma and evidence of a *VHL* gene mutation but no other stigmata of VHL.

22. **c. *SDHB* mutation.** Patients with multiple endocrine neoplasia (types 2A and 2B) possess a mutation in the *RET* proto-oncogene. Approximately half of these patients develop pheochromocytoma, but only about 3% of those with pheochromocytoma exhibit malignant potential. Ten percent to 20% of patients with VHL develop pheochromocytomas, but only 5% of those with pheochromocytoma have malignant disease. Pheochromocytoma among patients with neurofibromatosis type 1 is rare (1%), but malignant disease can be seen in more than 10%. Familial paraganglioma syndrome type 4 (*SDHB* mutation) carries the highest risk of malignancy (30% to 50%) among patients with the condition who develop pheochromocytoma (≈20%). Its pathologic cousin, familial paraganglioma syndrome type 1 (*SDHD* mutation), carries a negligible risk (<3%) of malignancy among patients who develop pheochromocytomas (≈20%).
23. **b. Plasma-free metanephrines or fractionated urinary metanephrines.** Methylated metabolites of catecholamines are known as *metanephrines.* Therefore normetanephrine (from norepinephrine) and metanephrine (from epinephrine) are collectively known as *metanephrines.* The vast majority of methylation occurs within the adrenal medulla or pheochromocytoma, when present. **Because this conversion of catecholamines to metanephrines is an uninterrupted process within pheochromocytomas, testing for these compounds is a much more sensitive means of tumor detection than the measurement of catecholamine levels, which may be paroxysmal.** Furthermore, measurement of levels of metanephrines is rather specific. Controversy exists regarding whether measurement of plasma-free metanephrines versus fractionated urinary metanephrine should be used as the initial test. The term “free” indicates that the metanephrines being measured are not conjugated by a sulfate moiety, whereas the term

"fractionated" simply indicates that normetanephrine and metanephrine levels are reported as separate values.

24. **c. To control tachycardia and arrhythmias that can result upon initiation of α blockade. Beta-blockade should never be started prior to appropriate α blockade.** In the absence of α blockade, β antagonists cause a potentiation of the action of epinephrine on the α_1 receptor, due to blockade of the arteriolar dilation at the β_2 receptor. Nevertheless, β blockade is at times necessary to control reflex tachycardia and arrhythmias that can result from α blockade.
25. **e. Priapism.** Patients with adrenal crisis are easily misdiagnosed with an acute abdomen. Children can exhibit hypoglycemic seizures. Persistent painful erections are generally not associated with adrenal insufficiency.
26. **c. Aldosteronoma.** All listed lesions, other than an aldosterone-producing adenoma, can develop outside of the adrenal gland.
27. **e. Vasoactive intestinal polypeptide (VIP). Ganglioneuromas are rare lesions that can arise in the adrenal glands and can secrete VIP, causing profound diarrhea in some patients.** Nevertheless, most ganglioneuromas are asymptomatic.
28. **c. 7%.** In a meta-analysis accounting for 515 adrenal cysts, the incidence of associated adrenal malignancy was 7%.
29. **a. Has a more favorable 5-year survival rate compared with adults.** The 5-year survival rate in children with adrenal cortical carcinoma is 54% compared with only 20% to 47% in adults.
30. **d. Cortisol.** Up to 74% of functional adrenocortical tumors produce excess cortisol.
31. **e. Oncocytic features.** The Weiss criteria should be applied with caution in pediatric cases and in those with oncocytic features.
32. **b. Mitotane alone or in combination with additional cytotoxic agents.** Mitotane has adrenolytic activity and is the first-line agent of choice in patients with metastatic adrenocortical carcinoma. The addition of streptozotocin or etoposide, doxorubicin, and cisplatin to mitotane has been shown to improve progression free survival compared to the use of mitotane alone.

33. **b. Complete surgical resection offers the best chance of cure.** Adjuvant radiation therapy has demonstrated a decreased rate of local recurrence but has not been shown to improve overall survival. Adjuvant mitotane therapy has been shown to significantly improve recurrence-free and overall survival. A tumor's functional status has not been consistently demonstrated to impact survival.
34. **b. Women taking oral contraceptives.** The urologist must be aware that the LDDST can yield as high as a 50% false-positive rate in women using oral contraceptives, because the contraceptives increase total (but not bioavailable) cortisol levels by raising the patient's cortisol-binding globulin concentrations.
35. **b. False.** The classic description by Robson in the 1950s suggested that radical nephrectomy should include adrenalectomy. Today, however, adrenalectomy is believed to be necessary only for large (T2) upper pole tumors, in cases in which an abnormality in the gland can be seen on preoperative imaging and in cases in which a vein thrombus is present to the level of the adrenal gland.
36. **c. Late-night salivary cortisol test, plasma aldosterone concentration, plasma renin activity, and plasma-free metanephrines. All patients presenting with an adrenal mass should be evaluated for cortisol and catecholamine hypersecretion.** Given the patient's history of hypertension, evaluation of primary aldosteronism should also be undertaken. Choice c is the best answer.
37. **e. Assessment of the adrenal tumor's functional status.** Although there is a high probability that the adrenal lesion, in this case, represents a recurrence of the patient's renal cell carcinoma, the functional status of the adrenal mass should be assessed. A pheochromocytoma may always be lurking.
38. **b. Counseling for left adrenalectomy.** Both the right and left cortisol gradients suggest proper catheter placement for adrenal vein sampling. The aldosterone ratio of 2.5:1 demonstrates left lateralization of autonomous aldosterone secretion, and counseling for left adrenalectomy is appropriate.

39. **e. All of the above.** The patient is less than 50 years old; therefore genetic screening is recommended. Up to 25% of patients who appear to have sporadic pheochromocytoma on presentation turn out to have germline mutations upon genetic testing. Repeat metabolic testing at 2 weeks after resection is prudent. Most experts recommend additional biochemical testing at 6 months, followed by annual lifelong screening. More than 15% of patients will demonstrate recurrence of pheochromocytoma in the first 10 years after a successful resection. Recurrent disease has been reported more than 15 years following adrenalectomy; therefore lifelong annual biochemical screening is advised. Although cross-sectional imaging is not absolutely required in the face of a negative biochemical workup, most surgeons obtain at least one study at some point during the postoperative follow-up.

Pathology

1. **a. No additional therapy is indicated because this is a benign tumor.** This is a benign myolipoma. Notice that it consists of fat mixed with hematopoietic elements.
2. **d. Short-term follow-up with imaging.** The marked nuclear variability, increased mitotic figures, and positive staining for Ki-67 all strongly suggest adrenal corticocarcinoma. No further pathologic information is necessary. These patients have a high risk of developing metastatic disease, at which time mitotane would be considered.

Imaging

1. **a. Adrenal adenoma.** Adrenal nodules that are less than 10 HU in density are almost always benign nodules, most often adrenal adenomas.

Chapter Review

1. The right adrenal gland is triangular in shape; the left is crescent shaped.
2. The zona glomerulosa secretes mineralocorticoids, the zona fasciculata secretes glucocorticoids, and the zona reticularis secretes sex steroids.
3. The production of cortisol is circadian, with the peak occurring in the early morning and the nadir at 11:00 PM.
4. Adrenal androgens are under the control of ACTH.
5. Ectopic ACTH production almost always originates from malignant tissue.
6. Renin release is stimulated by low renal perfusion pressure, increased renal sympathetic nervous activity, and low sodium.
7. Mineralocorticoid production results in sodium retention and volume expansion initially; however, with continued production, the kidney escapes from the sodium-retentive action of the hormone.
8. Sodium loading reduces endogenous aldosterone and renin production in those patients who do not have autonomous aldosterone secretion from aldosterone producing tumors.
9. The predictors of persistent hypertension following adrenalectomy for primary aldosteronism include (1) age older than 50 years, (2) the requirement for more than two antihypertensive agents preoperatively, (3) a first-degree relative with hypertension, (4) prolonged duration of hypertension prior to adrenalectomy, and (5) renal insufficiency.
10. Ki-67 staining of adrenal tissue is perhaps the best indicator of malignancy.
11. Chromogranin A elevation in the serum has been used as a confirmatory test in patients with pheochromocytoma.
12. Restoration of intravascular volume is the most important component of preoperative preparation in patients with pheochromocytoma.
13. The most frequent cause of adrenal insufficiency in the United States is autoimmune adrenalitis; in developing countries, it is tuberculosis.
14. Patients with congenital adrenal hyperplasia have a high risk for developing benign adrenal corticoadenomas.
15. The Weiss criteria distinguish benign from malignant adrenal tumors. The presence of three or more of the Weiss criteria is associated with

malignancy. When oncocytic features are present, the criteria should be used with caution.

16. An attenuation of less than 10 Hounsfield units on unenhanced computed tomography scan is strongly suggestive of an adrenal adenoma.
17. There are four histologic types of adrenal cysts: (1) pseudocyst, (2) endothelial, (3) epithelial, and (4) parasitic.
18. Positron emission tomography (PET) scan is the preferred imaging modality for pheochromocytoma.
19. Pheochromocytomas recur in up to 16% of patients who have had a complete surgical resection; 50% of the recurrences are malignant.
20. Metastases to the adrenal are common.
21. Production of aldosterone in the zona glomerulosa is primarily regulated by angiotensin II through the renin-angiotensin-aldosterone system and potassium levels.
22. Epinephrine is virtually a unique product of the adrenal gland, and when it is the dominant catechol produced by a tumor, an adrenal origin is suggested.
23. Cushing disease is a result of an overproduction of ACTH by the pituitary. It accounts for some 70% of endogenous Cushing syndrome.
24. Urolithiasis is seen in up to 50% of patients with Cushing syndrome.
25. Up to 30% of patients with Cushing disease who undergo bilateral adrenalectomy may develop Nelson syndrome—progressive growth of the pituitary adenoma causing increased intracranial pressure and compression of the ocular chiasm.
26. Although hypokalemia has been classically described as a common finding in primary aldosteronism, only 9%–37% of newly diagnosed patients are hypokalemic.
27. Conversion of catecholamines to metanephrines is an uninterrupted process within pheochromocytomas; testing for these compounds is a much more sensitive means for tumor detection than the measurement of catecholamine levels, which may be paroxysmal.
28. Ganglioneuromas are rare lesions that can arise in the adrenal glands and can secrete the vasoactive intestinal polypeptide, causing profound diarrhea in some patients.

29. All patients presenting with an adrenal mass should be evaluated for cortisol and catecholamine hypersecretion.

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Surgery of the Adrenal Glands

Sey Kiat Lim, and Koon Ho Rha

Keywords

adrenalectomy; adrenal gland; ablative therapy; laparoscopic; open; operative techniques; partial adrenalectomy; robot-assisted; surgery

Questions

1. Which of the following statements regarding the anatomy of the right adrenal gland is TRUE?
    a. Its vein drains into the right renal vein.
    b. It is usually located on the upper pole and lateral border of the right kidney.
    c. The medulla is innervated by preganglionic sympathetic nerve fibers.
    d. It is usually located more caudal than the left adrenal gland and is closely related to the diaphragm.
    e. The right adrenal gland only receives arterial supply from the right renal artery.
2. What is the risk of malignancy in an incidentaloma of more than 6 cm?
    a. 2%
    b. 6%
    c. 10%
    d. 25%

e. 50%

3. Absolute contraindications to laparoscopic adrenalectomy include:
    a. significant abdominal adhesions.
    b. adrenal mass greater than 12 cm in size.
    c. invasive adrenal cortical carcinoma with thrombus in the inferior vena cava.
    d. malignant pheochromocytoma.
    e. all of the above.
4. Which of the following statements regarding the perioperative management of pheochromocytoma is FALSE?
    a. Preoperative sympatholytic therapy with α-adrenergic blockers should be started for at least 2 weeks before surgery.
    b. Phenoxybenzamine may lead to tachycardia, and β-adrenergic blockade should be started before phenoxybenzamine administration.
    c. Vasopressins and aggressive fluid infusion are often necessary after ligation of the adrenal vein or excision of a pheochromocytoma.
    d. The adrenal vein should be ligated early during surgical excision of a pheochromocytoma.
    e. Intravenous drugs with short half-lives are preferred versus longer-acting drugs for controlling blood pressure fluctuations during surgical excision of a pheochromocytoma.
5. The lumbodorsal posterior approach to open adrenalectomy is not ideal in any of the following circumstances EXCEPT:
    a. large adrenal tumors.
    b. bilateral adrenal hyperplasia.
    c. bilateral small adrenocortical carcinoma.
    d. patients with ventilatory difficulties.
    e. b and c.
6. In bilateral adrenalectomy, steroids replacement should be started:
    a. on induction of general anesthesia.
    b. after ligation of the right adrenal vein.
    c. after excision of both adrenal glands.
    d. during closing of the abdominal incision.
    e. in the recovery room after surgery.

7. A 57-year-old male with no significant medical problems presented with a right-sided abdominal mass. Computed tomographic (CT) imaging showed an 18-cm right adrenal tumor with invasion of the upper pole of the right kidney and tumor thrombus extending into the retrohepatic inferior vena cava. Which is the best surgical approach for this patient?
    a. Open lumbodorsal posterior approach
    b. Open anterior transabdominal approach
    c. Open thoracoabdominal approach
    d. Laparoscopic transperitoneal approach
    e. Robot-assisted laparoscopic transperitoneal approach
8. Which of the following is NOT a possible approach to the left adrenal gland in the anterior transabdominal approach?
    a. Through the gastrocolic ligament
    b. Through the lienorenal ligament
    c. Through the transverse mesocolon
    d. Through the lesser omentum
    e. All the above are possible approaches
9. Which of the following statements on partial adrenalectomy is FALSE?
    a. The adrenal gland is usually exposed but not mobilized.
    b. The arterial supply of the adrenal gland can be ligated without devascularizing the gland.
    c. It is generally safe to ligate the main adrenal vein if the adrenal gland is not mobilized.
    d. A repeat partial adrenalectomy is indicated in a small local recurrence of an adrenal mass previously treated with partial adrenalectomy.
    e. Partial adrenalectomy may be indicated in multiple endocrine neoplasia type IIA.
10. During a right adrenalectomy, severe bleeding is encountered. The possible causes include all of the following EXCEPT:
    a. right adrenal vein avulsion at the origin on the inferior vena cava.
    b. avulsion of the right hepatic vein branch.
    c. disruption of the adrenal capsule.

d. inadvertent ligation of the upper pole renal artery.
e. all of the above.

11. Which of the following is not an advantage of retroperitoneal adrenalectomy when compared to transperitoneal approach?
    a. Reducing the risk of injury to bowels and abdominal organs
    b. Larger working space and clearer anatomical landmarks
    c. Lower incidence of hemodynamic and respiratory morbidities
    d. Usually the approach of choice in patients with multiple previous abdominal surgeries
    e. All of the above are advantages of the retroperitoneal approach over the transperitoneal approach
12. With regard to intraoperative complications during adrenalectomy, which of the following is NOT the appropriate management?
    a. In an event when splenectomy is performed for uncontrollable splenic hemorrhage, pneumococcus, *Haemophilus influenzae* type B (Hib), and meningococcus vaccinations should be given intraoperatively or immediately postoperatively.
    b. Postoperative drainage high in triglycerides and amylase is indicative of pancreas injury. Management consists of bowel rest with parenteral nutrition.
    c. Small diaphragmatic injuries if recognized intraoperatively should be repaired and a chest drain inserted.
    d. Inadvertent ligation of the superior mesenteric vein should be managed with revascularization and bowel resection.
    e. All of the above are appropriate management.

Answers

1. **c. The medulla is innervated by preganglionic sympathetic nerve fibers. The right adrenal vein drains directly into the inferior vena cava and derives its arterial supply through the superior, middle, and inferior adrenal arteries from the inferior phrenic artery, the abdominal aorta, and the renal artery, respectively.** It is usually more cephalad than the left adrenal gland and is located at the upper pole and medial border of the right

kidney. It is closely related to the diaphragm posteriorly, and thus extra care should be taken during surgical dissection, especially of large tumors.

2. **d. 25%. Of adrenal lesions larger than 6 cm, 25% are adrenal cortical carcinomas, and these larger lesions should be resected. Risk of malignancy in lesions smaller than 4 cm is 2%.** Approximately 6% of adrenal lesions between 4 and 6 cm are malignant, and surgical resection can be considered in appropriate individuals.
3. **c. Invasive adrenal cortical carcinoma with thrombus in the inferior vena cava.** Previous abdominal surgeries may result in dense intra-abdominal adhesions that may make surgery difficult. With improved laparoscopic techniques and equipment, large adrenal tumors are no longer considered absolute contraindication to the laparoscopic approach. Literature had described the feasibility of resecting large adrenal tumors laparoscopically. However, the risk of open conversion is greater in such large tumors. Invasion of adrenal cortical carcinoma into surrounding structures will make laparoscopic resection very difficult, and extension of tumor into the inferior vena cava should be considered an absolute contraindication to the laparoscopic approach.
4. **b. Phenoxybenzamine may lead to tachycardia, and β-adrenergic blockade should be started before phenoxybenzamine administration.** Preoperative sympatholytic therapy with α-adrenergic blockers for at least 2 weeks before surgery helps in both hemodynamic and glucose control and should be continued until the day of surgery. Phenoxybenzamine, being nonselective, may lead to tachycardia, and β-adrenergic blockade may be necessary. Because phenoxybenzamine is an irreversible noncompetitive α-adrenergic blocker, prolonged hypotension in the immediate postoperative period and central nervous system effects such as somnolence may be expected. **β-Adrenergic blockade, if needed, must be given with caution in patients with myocardial depression and started only after phenoxybenzamine therapy.** Aggressive fluid management with volume repletion is necessary after removal of pheochromocytoma or ligation of the adrenal vein,

because hypotension can occur as a result of sudden loss of tonic vasoconstriction. The adrenal vein should be ligated early during the resection of pheochromocytoma to avoid the systemic release of catecholamines during manipulation of the adrenal gland. Lastly, drugs with rapid onset and short half-lives, such as nitroprusside, phentolamine, nitroglycerin, or nicardipine, are generally preferred in intraoperative hypertensive episodes, because hypotension can occur after ligation of the adrenal vein or excision of the pheochromocytoma.

5. **b. Bilateral adrenal hyperplasia.** Although the posterior approach is the most direct route to the adrenal glands and no major muscles are divided, surgical exposure is limited. In addition, access to the adrenal vein and great vessels is more difficult, which may be problematic in event of excessive intraoperative bleeding. Thus this approach is not ideal in large adrenal tumors or adrenocortical carcinoma. **The prone position will also make ventilating the patient difficult. However, this approach provides ready access to both adrenal glands through two separate incisions and is ideal for bilateral adrenal hyperplasia or small benign adrenal tumors.**
6. **c. After excision of both adrenal glands.** It is generally recommended that steroid replacement be started as soon as both adrenal glands are excised to minimize acute adrenal insufficiency. Patients can present with back/abdominal pain, nausea, vomiting, diarrhea, hypotension, fever, hypoglycemia, and hyperkalemia.
7. **c. Open thoracoabdominal approach.** Surgical exposure is limited with the lumbodorsal posterior approach, and this approach should not be used for large tumors or adrenocortical carcinoma. Although the open anterior transabdominal approach using the subcostal or chevron incision gives a fairly good and adequate exposure for most cases, this approach might not be adequate in this case in view of the involvement of the retrohepatic inferior vena cava. The open thoracoabdominal approach is generally reserved for large and invasive tumors with extensive involvement of surrounding structures or vena cava that cannot be safely removed via the anterior transabdominal approach. The thoracoabdominal approach

is particularly useful in right-sided tumors, because the liver and inferior vena cava can limit exposure, whereas on the left side, the spleen and pancreas can generally be elevated to provide adequate exposure. Minimally invasive approaches such as conventional laparoscopic or robot-assisted laparoscopic approach are usually contraindicated in such large tumors with extensive involvement of the great vessels.

8. **e. All the above are possible approaches.**
9. **d. A repeat partial adrenalectomy is indicated in a small local recurrence of an adrenal mass previously treated with partial adrenalectomy.** The arterial supply of the adrenal gland forms a plexus circumferentially around the gland and can usually be taken without fear of devascularizing the adrenal cortex, and the gland will remain viable as long it remains attached to the kidney or to an area of unmobilized connective tissue. **The venous system drains into a central adrenal vein, and the main adrenal vein can be taken as long as the remnant adrenal gland remains in situ without mobilization. However, it would be prudent to preserve the main adrenal vein as long as it is safe and adequate margins can be obtained.** A local recurrence after a previous partial adrenalectomy, regardless of size of recurrence, is an absolute contraindication to a repeat partial adrenalectomy. Multiple endocrine neoplasia type IIA is associated with adrenal tumors, and therefore partial adrenalectomy may be indicated in these patients.
10. **d. Inadvertent ligation of the upper pole renal artery.** All of the vascular injuries noted cause bleeding except ligation of an upper pole renal artery. Ligation of this artery would result in upper pole devascularization, not hemorrhage.
11. **b. Larger working space and clearer anatomical landmarks.** The main advantage of the retroperitoneal approach is that entry into the peritoneum is avoided, and thus complications such as visceral and bowel injuries are minimized. In the absence of pneumoperitoneum, hemodynamic and respiratory morbidities are also reduced. In addition, dense intraperitoneal adhesions arising from previous surgery or inflammation are averted by operating in the retroperitoneum. The main disadvantage of the retroperitoneal

approach is the limited working space that makes dissection of large tumors difficult. Finally, the lack of anatomic landmarks and the abundant retroperitoneal adipose tissues may pose a significant challenge to surgeons inexperienced with the retroperitoneum. The main advantage of the lateral approach over the posterior approach is the ease of conversion into the transperitoneal approach should difficulties be encountered. In contrast, the prone retroperitoneal approach allows for bilateral adrenalectomy without patient repositioning.

12. **a. In an event when splenectomy is performed for uncontrollable splenic hemorrhage, pneumococcus, *Haemophilus influenzae* type B (Hib), and meningococcus vaccinations should be given intraoperatively or immediately postoperatively.** Vaccinations should be given only after patients have recovered adequately from their surgeries (usually just before discharge from hospital) and not intraoperatively or immediately postoperatively. Diaphragmatic injuries, even if small, should be repaired due to the risk of abdominal content herniation and bowel strangulations. Ligation of the superior mesenteric vein is potentially fatal due to large segments of bowel congestion. Management must include revascularization and resection of unhealthy segment of bowels if necessary.

Chapter Review

1. The right adrenal vein enters the inferior vena cava in a posterior lateral position. When torn, the vena cava must be rotated to gain access to suture the defect.
2. Local regional recurrence in the adrenal cortical carcinoma occurs in 60% of cases.
3. It is important to note that the tail of the pancreas can lie adjacent to the upper pole of the left kidney and adrenal. Care must be taken not to injure this organ.
4. In a thoracoabdominal incision, the diaphragm should not be incised radially but, rather, circumferentially because the former results in a

phrenic nerve injury with an atonic diaphragm lateral to the incision.

5. An Addisonian crisis is most commonly seen after excision of an adrenal tumor that secretes cortisol as the contralateral adrenal is suppressed.
6. In removing large adrenal masses, it is important to be careful not to ligate an upper pole renal artery branch because this will result in an infarction of the renal segment that is served by this artery.
7. The most common sites from which metastases occur to the adrenal are lung, breast, kidney, and melanoma.
8. Before resecting aldosterone-secreting tumors, consideration should be given to preoperative administration of spironolactone and correction of hypokalemia and hypomagnesemia if they exist.
9. Twenty-five percent of adrenal lesions greater than 6 cm are adrenal cortical carcinomas, and these larger lesions should be resected. The risk of malignancy in lesions less than 4 cm is 2%.
10. Preoperative sympatholytic therapy with α-adrenergic blockers for at least 2 weeks before surgery in patients with a pheochromocytoma is necessary because it restores vascular volume and helps in both hemodynamic and glucose control. If a β blocker is necessary to control cardiac arrhythmias, it should be begun after full alpha blockade has been achieved.

PART XIII

Urine Transport, Storage, and Emptying

OUTLINE

108. Surgical, Radiographic, and Endoscopic Anatomy of the Female Pelvis

109. Surgical, Radiographic, and Endoscopic Anatomy of the Male Pelvis

110. Physiology and Pharmacology of the Bladder and Urethra

111. Pathophysiology and Classification of Lower Urinary Tract Dysfunction: Overview

112. Evaluation and Management of Women With Urinary Incontinence and Pelvic Prolapse

113. Evaluation and Management of Men With Urinary Incontinence

114. Urodynamic and Video-Urodynamic Evaluation of the Lower Urinary Tract

115. Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

116. Neuromuscular Dysfunction of the Lower Urinary Tract

117. Overactive Bladder

118. The Underactive Detrusor

119. Nocturia

120. Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

121. Conservative Management of Urinary Incontinence: Behavioral and Pelvic Floor Therapy, Urethral and Pelvic Devices

122. Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

123. Retropubic Suspension Surgery for Incontinence in Women

124. Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

125. Slings: Autologous, Biologic, Synthetic, and Midurethral

126. Complications Related to the Use of Mesh and Their Repair

127. Additional Therapies for Storage and Emptying Failure

128. Aging and Geriatric Urology

129. Urinary Tract Fistulae

130. Bladder and Female Urethral Diverticula

131. Surgical Procedures for Sphincteric Incontinence in the Male

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Surgical, Radiographic, and Endoscopic Anatomy of the Female Pelvis

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Keywords

anatomy; endoscopic anatomy; female pelvis; radiographic anatomy; surgical anatomy

Questions

1. The uterine vessels are found _____ to the ureter.
    a. laterally
    b. posteriorly
    c. anteriorly
    d. medially
    e. running together in a common sheath
2. All of the following are TRUE EXCEPT:
    a. The angle between the inferior pubic rami is obtuse in women.
    b. The female pelvis is larger and broader than the male pelvis.
    c. The female inlet is heart shaped (as compared with the oval-shaped male inlet).
    d. The female pelvis is shorter and wider, with a less pronounced sacral promontory.

e. Female pelvic inlet faces anteriorly, allowing pressure of the intra-abdominal and intrapelvic contents to be directed to the bony pelvis.

3. In contrast to that of the male, the female bladder neck:
    a. has extensive adrenergic innervation.
    b. has a thickened middle smooth muscle layer.
    c. is largely responsible for urinary continence.
    d. is surrounded by type I (slow-twitch) fibers.
    e. has longitudinal smooth muscle fibers that extend to the external meatus.
4. Relative to the anterior portion of the uterosacral ligament, the ureter:
    a. lies inferior.
    b. lies posterior.
    c. lies superior.
    d. lies lateral.
    e. runs along with it.
5. To avoid denervation of the striated urethral sphincter, incisions through the vaginal wall to enter the retropubic space should be made:
    a. perpendicular to the urethra.
    b. over the urethra.
    c. close to the lateral margins of the urethra.
    d. cephalad to the bladder neck.
    e. far lateral in the vaginal wall, parallel to the urethra.
6. The levator ani muscles make up the largest portion of the pelvic cavity floor. This consists of the:
    a. iliopsoas, puborectalis, and obturator internus.
    b. obturator externus, psoas major, and psoas minor.
    c. coccygeus, pubococcygeus, and iliopsoas.
    d. iliococcygeus, puborectalis, and pubococcygeus.
    e. obturator internus, pubococcygeus, and psoas minor.
7. A Maritus (labial fat pad) rotational flap used in the repair of vesicovaginal fistula receives blood supply from the:
    a. terminal branches of the internal pudendal artery and vein.
    b. superficial inferior epigastric vessels.

c. inferior epigastric vessels.
d. accessory pudendal vessels.
e. external pudendal vessels.

8. Translabial ultrasound is valuable because it does not distort or compress the tissue. It is commonly used to visualize:
   a. transvaginal mesh.
   b. urethral bulking agents.
   c. urethral implants.
   d. bladder wall thickness.
   e. all of the above.
9. The round ligament:
   a. terminates in the uterus.
   b. is the main source of blood supply to the ovaries.
   c. terminates in the labia.
   d. is the male homolog of the spermatic cord.
   e. is NOT part of the broad ligament.
10. All of the following are true regarding the perineal bodyEXCEPT:
   a. Damage to the perineal body during labor can cause damage to fibers of the internal anal sphincter.
   b. It is the convergence of the bulbospongiosus, external anal sphincter and superficial and deep transverse perineal muscles.
   c. Perineal body attaches to the posterior border of the perineal membrane.
   d. Perineal body is made up of muscle and collagenous and elastic fibers and lies at the central point of the perineum.
   e. Episiotomy is angled laterally to avoid damage to fibers of the external anal sphincter.
11. The transversalis fascia is:
   a. part of the inner stratum.
   b. part of the outer stratum.
   c. part of the intermediate stratum.
   d. continuous with the endopelvic fascia.
   e. b and d.
12. After a sacrospinous ligament fixation, the patient wakes up with pain in the posterior and lower leg. The nerve that was likely compromised is the:

a. femoral nerve.
b. pudendal nerve.
c. sacral plexus.
d. obturator nerve.
e. a and b.

13. The arcus tendineus fascia pelvis:
    a. is present in females.
    b. attaches from the ischial spine to the sacrum.
    c. is also known as the tendinous arc.
    d. is present only in males.
    e. all of the above.
14. Please identify the correct pair(s) of arterial supply and lymphatic drainage of the vagina.
    a. Uterine arteries: superficial inguinal nodes
    b. Internal pudendal artery: sacral common iliac and superficial inguinal nodes
    c. Vaginal arteries: external iliac nodes
    d. Uterine arteries: sacral common iliac nodes
    e. Internal pudendal artery: external iliac nodes
15. Squamous metaplasia of the bladder:
    a. is a premalignant lesion.
    b. is a sign of an underlying serious infection.
    c. should always be treated with surgical resection.
    d. is a normal finding in premenopausal females.
    e. is none of the above.
16. The levator ani attaches to all of the following EXCEPT:
    a. perineal body.
    b. pubis.
    c. coccyx.
    d. vagina.
    e. arcus tendineus fascia pelvis.
17. The ureter can be injured during a hysterectomy:
    a. at the time of the division of the ovarian artery.
    b. at the time of the division of the uterine artery.
    c. at the time of the division of the cardinal artery.
    d. while dissecting out the cervix.

e. all of the above.

18. During a sacrospinous ligament vault suspension, significant bleeding is encountered. The bleeding vessels most likely originate from the:
    a. anterior division of the internal iliac artery.
    b. posterior division of the internal iliac artery.
    c. perineal artery.
    d. mesenteric artery.
    e. epigastric artery.
19. The peritoneum may be accessed transvaginally through the:
    a. posterior fornix.
    b. anterior fornix.
    c. rectovaginal septum.
    d. lateral fornices.
    e. vesicovaginal space.
20. The cavernous nerve, responsible for clitoral engorgement during sexual activity, is derived from:
    a. the pudendal nerve.
    b. the dorsal nerve of the clitoris.
    c. the superior hypogastric plexus.
    d. the inferior hypogastric plexus.
    e. the obturator nerve

Answers

1. **c. Anteriorly.** In women, the ureter first runs posterior to the ovary, then turns medially to run deep to the base of the broad ligament before entering a loose connective tissue tunnel through the substance of the cardinal ligament.
2. **c. The female inlet is heart shaped (as compared with the oval-shaped male inlet).** The female inlet is oval in shape (as compared with the heart-shaped male inlet) and wider, which contributes to the weakness of the pelvic floor.
3. **e. Has longitudinal smooth muscle fibers that extend to the external meatus.** At the female bladder neck, the inner

longitudinal fibers converge radially to pass downward as the inner longitudinal layer of the urethra. The middle circular layer does not appear to be as robust as that of the male. The female bladder neck differs strikingly from the male in possessing little adrenergic innervation.

4. **d. Lies lateral.** The uterosacral originates from the greater sciatic foramen and inserts into the lateral aspect of the fascia surrounding the cervix, isthmus of the uterus, and vaginal wall. The ureter lies lateral to the anterior portion of the uterosacral ligament (closest to the cervix).
5. **e. Far lateral in the vaginal wall, parallel to the urethra.** Somatic and autonomic nerves to the urethra travel on the lateral walls of the vagina near the urethra. During transvaginal incontinence surgery, the anterior vaginal wall should be incised laterally to avoid these nerves and prevent type III urinary incontinence.
6. **d. Iliococcygeus, puborectalis, and pubococcygeus.** This is innervated by the fourth sacral nerve. The name of each component is derived from their attachments.
7. **e. External pudendal vessels.** The labial fat pads received blood supply from the external pudendal branches of the femoral vessels.
8. **e. All of the above.**
9. **c. Terminates in the labia.** The round ligament exits at the internal inguinal ring and crosses over the external iliac artery, terminating in the mons pubis of the labia. The male homolog of the round ligament is the gubernaculum.
10. **a.** Damage to the perineal body during labor can cause damage to fibers of the internal anal sphincter.
11. **e. b and c.** The transversalis fascia is part of the outer stratum and is continuous with the endopelvic and lateral pelvic fascia. Both the transversalis and endopelvic fascia play important roles at the exit points of the pelvic organs. The endopelvic fascia extends from the uterine artery down to where the vagina and levator ani fuse.
12. **c. Sacral plexus.** During an sacrospinous ligament fixation, the sacral plexus is in jeopardy of being injured because it lies immediately posterior to the sacrospinous ligament as it leaves

through the greater sciatic foramen. The pudendal nerve may also be injured during this repair.

13. **c. Is also known as the tendinous arc.** The arcus tendineus fascia pelvis, or tendinous arc, is a thickened band of the pelvic fascia that runs from the ischial spine to the pubic bone. It originates from the pubic bone laterally and is connected to the pubovesical ligament medially and the tendinous arch of the levator ani.
14. **b. Internal pudendal artery: sacral, common iliac, and superficial inguinal nodes.** The superior part: blood supply—uterine arteries/drainage—internal and external iliac nodes middle and inferior: blood supply—vaginal arteries/drainage—internal and external iliac nodes Inferior vagina: blood supply—internal pudendal artery/drainage—sacral, common iliac, and superficial inguinal nodes.
15. **d. Is a normal finding in premenopausal females.** Premenopausal women can have normal signs of squamous metaplasia at the trigone and base of the bladder. This is a nonkeratinizing metaplasia or vaginal metaplasia that is hormonally responsive and is a normal variant.
16. **e. Arcus tendineus fascia pelvis.** The tendinous arc of the levator ani serves as the origin of the muscles of the pelvis diaphragm: pubococcygeus and iliococcygeus. The muscle bordering this hiatus has been referred to as "pubovisceral" because it provides a sling for (pubourethralis, puborectalis), inserts directly into (pubovaginalis, puboanalis, levator prostate), or inserts into a structure intimately associated with the pelvic viscera. The coccygeus muscle extends from the sacrospinous ligament to the lateral border of the sacrum and coccyx to complete the pelvic diaphragm.
17. **e. All of the above.** The ureter is vulnerable to injury because it comes in close proximity to many of the structures that are dissected and divided during a radical hysterectomy. It crosses the infundibulopelvic ligament under the ovarian artery and is just medial to the uterine artery. It also passes through the cardinal ligament and lies in close proximity to the cervix.
18. **b. Posterior division of the internal iliac artery.**

19. **a. Posterior fornix.** Because the apex of the vagina is covered with the peritoneum of the rectouterine pouch, the peritoneal cavity may be accessed through the posterior fornix.
20. **d. The inferior hypogastric plexus.**

Chapter Review

1. The pelvic inlet faces anteriorly, allowing pressure of the intra-abdominal and intrapelvic contents to be directed to the bony pelvis, rather than the muscles and the fascia. The bony pelvis should be visualized in the supine position.
2. The arcus tendinous fascia pelvis (ATFP) and arcus tendinous levator ani (ATLA) are two distinct structures, serving as key players in fascial attachment. ATFP is the main point for fascial attachment. The levator ani muscles attach to the ATFP and cover the obturator internus.
3. The ischial spine and sacrospinous ligament are important landmarks during vault suspension to avoid injury to the sacral plexus and pudendal nerve.
4. The sustained resting tone of the pelvic floor muscles (pubococcygeus, puborectalis, and iliococcygeus) support pelvic viscera, resists increases in intra-abdominal pressure, and is crucial in passive control of urinary and fecal continence.
5. Translabial ultrasound is useful for visualization of transvaginal mesh, urethral bulking agents, and urethral implants.
6. Magnetic resonance imaging (MRI) is the most comprehensive and conclusive imaging modality available to image the female pelvis without the use of radiation.
7. T2-weighted MRI is the best imaging technique to visualize and localize urethral diverticula and to differentiate them from other benign vaginal masses.

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Surgical, Radiographic, and Endoscopic Anatomy of the Male Pelvis

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Keywords

anatomy; lower abdominal wall; male pelvis; male perineum

Questions

1. The greater and lesser sciatic foramina are separated by the:
    a. sacrotuberous ligament.
    b. Cooper (pectineal) ligament.
    c. arcuate line.
    d. sacrospinous ligament.
    e. piriformis muscle.
2. During inguinal incisions, the vessels invariably encountered in Camper fascia are the:
    a. superficial inferior epigastric artery and vein.
    b. superficial circumflex iliac artery and vein.
    c. external pudendal artery and vein.
    d. gonadal artery and veins.
    e. accessory obturator vein.
3. Rupture of the penile urethra at the junction of the penis and scrotum can result in urinary extravasation into all of the following

structures EXCEPT the:

a. anterior abdominal wall up to the clavicles.
b. scrotum.
c. penis, deep to the dartos fascia.
d. perineum in a “butterfly” pattern.
e. buttock.

4. Accessory obturator veins (from the external iliac artery) and accessory obturator arteries (from the inferior epigastric artery) are encountered in:
    a. 50% and 25% of patients, respectively.
    b. 5% and 50% of patients, respectively.
    c. 50% and 75% of patients, respectively.
    d. 25% and 50% of patients, respectively.
    e. 25% and 5% of patients, respectively.
5. A retractor blade has rested on the psoas muscle during a prolonged procedure, resulting in a femoral nerve palsy. Postoperatively, the patient will experience:
    a. inability to flex the hip and numbness over the anterior thigh.
    b. inability to flex the knee and numbness over the thigh.
    c. numbness over the anterior thigh only.
    d. inability to extend the knee and numbness over the anterior thigh.
    e. inability to flex the knee only.
6. Autonomic nerves contributing to the pelvic plexus include the:
    a. superior hypogastric nerves from the para-aortic plexuses.
    b. pelvic sympathetic trunks.
    c. pelvic parasympathetic neurons from the sacral spinal cord.
    d. a and c only.
    e. a, b, and c.
7. To preserve the vascular supply to the ureter, incisions in the peritoneum should be made:
    a. medially in the abdomen and laterally in the pelvis.
    b. laterally in the abdomen and medially in the pelvis.
    c. always medial to the ureter.
    d. always lateral to the ureter.
    e. directly over the ureter.

8. All of the following features of the ureterovesical junction cooperate to prevent vesicoureteral reflux EXCEPT:
    a. fixation of the ureter to the superficial trigone.
    b. sphincteric closure of the ureteral orifice.
    c. detrusor backing.
    d. telescoping of the bladder outward over the ureter.
    e. passive closure of the intramural ureter caused by bladder filling.
9. Which of the following statements about the trigone is TRUE?
    a. Epithelium is thicker than the rest of the bladder and densely adherent.
    b. Superficial smooth muscle is a continuation of Waldeyer sheath.
    c. Smooth muscle enlarges to form thick fascicles.
    d. Smooth muscle of the ureter forms the interureteric ridge (Mercier bar).
    e. When the bladder empties, the trigone is thrown into thick folds.
10. Arterial supply to the bladder includes:
    a. the superior vesical artery.
    b. the inferior vesical artery.
    c. the obturator artery.
    d. the inferior hemorrhoidal artery.
    e. all of the above.
11. Which of the following statements concerning the male striated urethral sphincter is TRUE?
    a. It is composed of type I (slow-twitch) and type II (fast-twitch) fibers.
    b. It is bounded above by the superior fascia.
    c. It receives motor blanches from the dorsal nerve of the penis.
    d. It is shaped like a signet ring and is 2 to 2.5 cm in length.
    e. It is densely supplied with proprioceptive muscle spindles.
12. The first branch of the pudendal nerve in the perineum is the:
    a. dorsal nerve of the penis.
    b. inferior rectal nerve(s).
    c. perineal nerve.

d. posterior femoral cutaneous branches.
e. posterior scrotal branches.

13. Lymphatic drainage from the scrotum travels:
    a. through perianal nodes to reach the pelvis.
    b. directly to the deep pelvic lymph nodes.
    c. through the superficial and deep inguinal lymph nodes.
    d. to prepubic nodes.
    e. to para-aortic lymph nodes along with testicular drainage.
14. Which layers of the scrotum and testicular tunics usually need to be debrided in patients with Fournier gangrene?
    a. The scrotal skin only
    b. The scrotal skin and dartos layer
    c. The scrotal skin, dartos layer, and external spermatic fascia
    d. The scrotal skin, dartos layer, and external cremasteric and internal spermatic fasciae, leaving the tunica vaginalis intact
    e. All tissues, including the tunica vaginalis
15. Lymphatic drainage from the bladder passes through the:
    a. external iliac lymph nodes.
    b. obturator and internal iliac lymph nodes.
    c. internal and common iliac lymph nodes.
    d. common iliac, periureteral, and para-aortic lymph nodes.
    e. a, b, and c.
16. To preserve potency during a radical cystectomy, ligation of the lateral and posterior vascular pedicles is best carried out:
    a. close to their origin from the internal iliac vessels.
    b. near the bladder.
    c. from beneath the bladder after rotating the prostate cephalad.
    d. as they cross the ureter.
    e. lateral to the rectum.
17. Which imaging modality/procedure would be optimal to assess degree of invasion of a bulbar urethral tumor?
    a. CT scan with contrast
    b. Retrograde urethrogram with fluoroscopy
    c. MRI of the pelvis with contrast with T1- and T2-weighted images
    d. Endoscopy with direct visualization

e. Ultrasound of the perineum

18. During transurethral resection of the prostate, what maneuver is most likely to result in urinary incontinence?
    a. Undermining of the floor of the bladder neck
    b. Deep resection distal to the verumontanum
    c. Superficial resection of the ejaculatory ducts
    d. Deep resection of intravesical protrusion of the prostate
    e. Deep resection of the lateral lobes of the prostate proximal to the verumontanum

Answers

1. **d. Sacrospinous ligament.** The sacrospinous ligament separates the greater and lesser sciatic foramina.
2. **a. Superficial inferior epigastric artery and vein.** The superficial inferior epigastric vessels are encountered during inguinal incisions and can cause troublesome bleeding during placement of pelvic laparoscopic ports or in an inguinal incision.
3. **e. Buttock.** Blood and urine can accumulate in the scrotum and penis deep to the dartos fascia after an anterior urethral injury. In the perineum, their spread is limited by the fusions of Colles fascia to the ischiopubic rami laterally and to the posterior edge of the perineal membrane; the resulting hematoma is therefore butterfly shaped. These processes will not extend down the leg or into the buttock, but they can freely travel up the anterior abdominal wall deep to Scarpa fascia to the clavicles and around the flank to the back.
4. **a. 50% and 25% of patients, respectively.** In half of patients, one or more accessory obturator veins drain into the underside of the external iliac vein and can easily be torn during lymphadenectomy. In 25% of people, an accessory obturator artery arises from the inferior epigastric artery and runs medial to the femoral vein to reach the obturator canal.
5. **d. Inability to extend the knee and numbness over the anterior thigh.** The femoral nerve (L2, L3, and L4) supplies sensation to the

anterior thigh and motor innervation to the extensors of the knee.

6. **e. a, b, and c.** The presynaptic sympathetic cell bodies reach the pelvic plexus by two pathways: (1) the superior hypogastric plexus and (2) the pelvic continuation of the sympathetic trunks. Presynaptic parasympathetic innervation arises from the intermediolateral cell column of the sacral cord.
7. **b. Laterally in the abdomen and medially in the pelvis. Blood supply to the pelvic ureter enters laterally; thus the pelvic peritoneum should be incised only medial to the ureter.**
8. **b. Sphincteric closure of the ureteral orifice.** The intravesical portion of the ureter lies immediately beneath the bladder urothelium and is therefore quite pliant; it is backed by a strong plate of detrusor muscle. With bladder filling, this arrangement is thought to result in passive occlusion of the ureter, like a flap valve.
9. **d. Smooth muscle of the ureter forms the interureteric ridge (Mercier bar).** Fibers from each ureter meet to form a triangular sheet of muscle that extends from the two ureteral orifices to the internal urethra meatus. The edges of this muscular sheet are thickened between the ureteral orifices (the interureteric crest, or Mercier bar) and between the ureters and the internal urethral meatus (Bell muscle).
10. **e. All of the above.** In addition to the vesical branches, the bladder may be supplied by any adjacent artery arising from the internal iliac artery.
11. **d. It is shaped like a signet ring and is 2 to 2.5 cm in length.** The membranous urethra spans on average 2 to 2.5 cm (range: 1.2 to 5 cm). In the male it is surrounded by the striated (external) urethral sphincter, which is often incorrectly depicted as a flat sheet of muscle sandwiched between two layers of fasciae. The striated sphincter is actually shaped like a signet ring, broad at its base and narrowing as it passes through the urogenital hiatus of the levator ani to meet the apex of the prostate.
12. **a. Dorsal nerve of the penis.** The pudendal nerve follows the vessels in their course through the perineum. Its first branch, the dorsal nerve of the penis, travels ventral to the main pudendal trunk in the Alcock canal.

13. **c. Through the superficial and deep inguinal lymph nodes.** The penis, scrotum, and perineum drain into the inguinal lymph nodes. These nodes can be divided into superficial groups and deep groups.
14. **b. The scrotal skin and dartos layer.** The external, cremasteric, and internal spermatic fasciae are embryologically distinct from the scrotal and dartos layers and have their own blood and nerve supplies. It is uncommon for them to be involved in the necrotic process in Fournier gangrene; therefore they can be spared. (In practice, in patients with Fournier gangrene, all scrotal tissue is debrided to the tunica vaginalis.)
15. **e. a, b, and c. In the bladder, the bulk of the lymphatic drainage passes to the external iliac lymph nodes. Some anterior and lateral drainage may go through the obturator and internal iliac nodes, and portions of the bladder base and trigone may drain into the internal and common iliac groups. Complete lymph node dissection during radical cystectomy should encompass all of these lymph node groups.**
16. **b. Near the bladder.** The bladder vasculature pierces the pelvic autonomic plexuses near the origin of the arteries from the internal iliac arteries. Ligation of these vessels proximally will injure the pelvic autonomic nervous plexuses. Ligation is best carried out near the bladder to avoid nerve damage.
17. **c. MRI of the pelvis with contrast with T1 and T2-weighted images.** MRI has superior soft tissue spatial resolution that is well suited to imaging of the pelvic floor and external genitalia structures. Tumors typically enhance on T1 imaging when gadolinium contrast is administered, and T2 images show organ anatomy and detail well, enabling invasion to be determined.
18. **b. Deep resection distal to the verumontanum.** The striated urethral sphincter is located just distal to the verumontanum, and its coaptation can be noted visually during cystourethroscopy. Deep resection in this area may damage the striated urethral sphincter and result in urinary incontinence.

Chapter Review

1. Scarpa fascia on the abdomen forms a distinct layer and is continuous with Colles fascia in the perineum medially. It fuses with the deep fascia of the thigh laterally. Colles fascia is continuous with the dartos fascia of the penis. Thus, urinary extravasation and infections confined by these fascia attachments do not extend down the legs or into the buttocks but may travel cephalad to the clavicles.
2. The internal oblique and the transversalis fascia fuse to form the conjoin tendon, which reinforces the posterior wall of the inguinal canal.
3. A direct hernia of the inguinal canal occurs medial to the inferior epigastric vessels; an indirect hernia occurs lateral to these vessels.
4. The superior vesicle artery arises from the proximal portion of the obliterated umbilical artery. The obliterated umbilical artery may be used to find the superior vesicle artery.
5. In 25% of people there is an accessory obturator artery that arises from the inferior epigastric artery and courses medial to the femoral vein.
6. The genitofemoral nerve courses along the ventral surface of the psoas muscle; the femoral nerve runs in the substance of the psoas muscle. Retractors compressing the muscle may result in nerve palsy. Sutures placed perpendicular to the muscle fibers may entrap the nerves.
7. The obturator nerve supplies the adductors of the thigh.
8. The male bladder neck receives abundant sympathetic innervation. The female bladder neck receives little sympathetic innervation.
9. The blood supply to the prostate enters at the 4 and 8 o'clock positions. It is important to understand this in prostatectomy when securing hemostasis.
10. Although lymphatic supply from the prostate is primarily to the obturator and internal iliac nodes, it may drain directly to the presacral and external iliac nodes.
11. Denonvilliers fascia separates the prostate from the rectum.

12. Scrotal lymphatics do not cross the median raphe; drainage is to the ipsilateral inguinal nodes. Lymphatics from the penis cross over extensively and may drain to either or both groin nodes irrespective of the side of the penis involved.
13. The cavernosal nerves pass by the tips of the seminal vesicles and lie within the leaves of the lateral endopelvic fascia and course very close to the apex of the prostate, where they are most vulnerable to injury.
14. The ureter is anterior (ventral) to the common iliac artery.
15. The blood supply to the pelvic ureter enters laterally.
16. The bladder is an intra-abdominal organ in the infant and may project above the umbilicus.
17. The muscle of the trigone has three layers: (1) the superficial layer, derived from the longitudinal muscle of the ureter; (2) the deep layer, which arises from Waldeyer sheath on the ureter and inserts at the bladder neck; and (3) the detrusor layer.
18. The bladder has a lateral pedicle that is lateral to the ureter and a posterior pedicle that is posteromedial to the ureter.

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Physiology and Pharmacology of the Bladder and Urethra

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Keywords

bladder pain; bladder; neurogenic bladder; neurourology; overactive bladder; pelvic floor; pharmacology; underactive bladder; urethra; urinary incontinence; uropharmacology; voiding dysfunction

Questions

1. The lower urinary tract is innervated by three sets of peripheral nerves. Which of the following is correct?
    a. Pelvic parasympathetic nerves arise at the sacral spinal cord level and excite the bladder and urethra.
    b. Sympathetic nerves inhibit the bladder body and excite the bladder neck and urethra.
    c. Pudendal (somatic) nerves excite the bladder body and the external urethral sphincter.
    d. All of the above are true.
    e. None of the above are true.
2. “Sensing” bladder volume is relevant during bladder storage. Which of the following is correct?
    a. It has been speculated the sense of imminent micturition arises in the urethra.

b. Bladder filling has been shown to correlate with episodic bursts of sensation and afferent discharge.
c. Nathan first described sensations of awareness during bladder filling.
d. Sensations of awareness during bladder distension can be mapped to the urinary bladder.
e. All are correct.

3. Bladder afferents can be:
a. characterized by responses to receptive field stimulation.
b. classified according to ability to respond to a diverse range of chemical mediators.
c. silent initially but sensitized during inflammatory processes.
d. variable in both morphology and function depending upon species.
e. all of the above.

4. Sensitization of afferents in bladder pathology can:
a. open ion channels in the nerve terminals.
b. release a number of inflammatory mediators.
c. develop rapidly and be relatively short lived.
d. be resolved easily.
e. a, b, and c only.

5. Patients with irritable bowel syndrome (IBS) often report changes in bladder function. Which of the following statements is correct?
a. This is an example of cross-organ sensitization.
b. In animal models, colonic inflammation rarely leads to bladder dysfunction.
c. Cross-organ sensitization only occurs between the gastrointestinal tract and the urinary bladder.
d. The mediators, which are responsible for these conditions, have been well described.
e. None are true.

6. In terms of nerves innervating the detrusor:
a. the majority express acetylcholinesterase enzyme.
b. acetylcholine and adenosine triphosphate (ATP) appear to provide the majority of the excitatory input.

c. release of both acetylcholine and ATP result in smooth muscle contraction.
d. additional substances released from efferent nerves include nitric oxide (NO) and vasoactive intestinal polypeptide.
e. all are correct.

7. In terms of adrenergic influences:
   a. reflex bladder activity can be modulated by α-1 adrenergic mechanisms.
   b. α-1 adrenergic mechanisms control blood pressure and tissue blood flow.
   c. the β-3 adrenergic receptors are present at a number of sites (both peripherally and centrally).
   d. β-3 receptor agonists, via effects on a number of sites, are a promising treatment for overactive bladder.
   e. all are correct.
8. Which of the following is TRUE about ATP?
   a. It is the main excitatory neurotransmitter for bladder contraction in humans.
   b. It can activate two main families of purinergic receptors: P2X and P2Y.
   c. Purinergic neurotransmission plays an important role in bladder overactivity and bladder pain.
   d. b and c are true.
   e. All are correct.
9. During bladder storage:
   a. bladder accommodation is dependent upon activation of sympathetic pathways.
   b. bladder accommodation is dependent upon quiescence of parasympathetic efferent pathways.
   c. intravesical pressure measurements are low when below the voiding threshold.
   d. the sympathetic reflex provides negative feedback.
   e. all are true.
10. During storage phase of the bladder:
   a. the urothelium plays an important role in accommodating urine storage.

b. the urothelium is only a barrier and exhibits no other functions.
c. increase of urothelial mediators during bladder filling can influence smooth muscle tone.
d. the urothelial surface cells change shape during bladder filling.
e. a, c, and d are correct.

11. The guarding reflex is a mechanism for maintaining continence and is characterized by:
    a. activation of pudendal motoneurons.
    b. increased outlet resistance.
    c. activation of external urethral sphincter motoneurons.
    d. activation of afferent input from the urethra or pelvic floor that leads to closure of the urethral outlet.
    e. all of the above.
12. Electrical stimulation of the sacral nerve roots is known to be:
    a. an effective treatment for refractory overactive bladder.
    b. an effective treatment for non-obstructive urinary retention.
    c. effective by modulation of central nervous system pathways.
    d. a better treatment compared with posterior tibial nerve stimulation.
    e. a, b, and c are correct.
13. In terms of bladder emptying:
    a. switching between bladder storage and emptying can occur involuntarily (reflex emptying) or voluntarily.
    b. reflex voiding only occurs in the normal adult.
    c. initial expulsion of urine consists of initial contraction of the urethral sphincter.
    d. relaxation of the urethral smooth muscle during micturition is achieved by release of acetylcholine.
    e. none are true.
14. The facilitatory urethra to bladder reflex was characterized by:
    a. Barrington.
    b. Delancey.
    c. de Groat.

d. increased urethral afferent activation promoting bladder emptying.
e. a and d are correct.

15. An essential control center for micturition in healthy individuals is:
    a. the dorsal pontine tegmentum.
    b. Barrington nucleus.
    c. the pontine micturition center (PMC).
    d. the M region.
    e. all of the above.
16. In terms of cerebral control of voiding in humans:
    a. coordinated relaxation and contraction of urethra and bladder is driven by a long-loop spinobulbospinal reflex.
    b. afferents activated during bladder filling synapse in the central periaqueductal gray (PAG) and PMC regions.
    c. spinobulbospinal voiding-reflex pathway functions as a "switch."
    d. absence of the "switch" would lead to incontinence.
    e. all of the above.
17. In terms of continence and brain–bladder control:
    a. it involves limbic areas associated with basic emotion and safety.
    b. it involves cortical circuits concerned with social propriety.
    c. the PAG and PMC form the main brainstem "switch."
    d. all of the above.
18. The contraction of detrusor by cholinergic muscarinic receptor agonists is characterized by:
    a. inositol triphosphate (ISP) hydrolysis and release of intracellular calcium.
    b. decreased calcium flux through nifedipine-sensitive calcium channels.
    c. involvement of the muscarinic M1 subtype only.
    d. stimulation of the Rho-kinase pathway.
    e. none of the above.
19. Stimulation of β-adrenergic receptors in human detrusor is characterized by:
    a. relaxation of detrusor smooth muscle.

b. involvement of β-2 and β-3 subtype receptors.
c. accumulation of cyclic adenosine monophosphate (cAMP).
d. all of the above.

20. Urethral tone and intraurethral pressure are influenced by:
    a. α adrenergic receptors.
    b. sympathetic innervation.
    c. number of intramural ganglia.
    d. parasympathetic innervation.
    e. a and b are correct.
21. Which transgenic animal models has (have) been published showing detrusor overactivity?
    a. Increased M3
    b. Decreased P2X3/P2X2
    c. Decreased BK
    d. Decreased β1-integrin
    e. c and d are correct
22. The lamina propria (LP) of the bladder is thought to be the critical compartment because of which cell(s) that can mediate interaction between urothelium and nerves within the LP?
    a. Fibroblasts
    b. Ganglion cells
    c. Myofibroblasts
    d. Interstitial cells
    e. c and d are correct
23. The primary mechanism of action of onabotulinumtoxinA is:
    a. blockade of intracellular vesicle fusion in presynaptic nerves.
    b. suppression of afferent nerves in the LP.
    c. blockade of M3 receptor on detrusor myocyte.
    d. opening of BK channel on detrusor myocyte.
    e. blockade of voltage-dependent calcium channels on detrusor myocyte.
24. What distinguishing feature(s) in the bladder urothelium distinguish(es) it from the LP, detrusor, and serosal compartments?
    a. Presence of connexin-43 (Cx43)
    b. Presence of purinergic receptors
    c. Presence of tight junction proteins

d. Presence of uroplakins
e. c and d are correct

25. During acute bacterial cystitis, uropathogenic bacteria induce a host immune response due to their interaction with which of the following receptors on the urothelial cell?
    a. TRPV1
    b. M2
    c. NK-A
    d. TLR4
    e. P2X3
26. Maintenance of normal bladder compliance during urinary storage is (are) due to:
    a. passive viscoelastic properties.
    b. active neural signaling.
    c. modulation of filling rate.
    d. all of the above.
    e. a and b are correct.
27. Cx43 is important in regulating detrusor contractility because:
    a. it regulates acetylcholine release by efferent nerves.
    b. it breaks down acetylcholine in the neuromuscular junction.
    c. it allows for passage of ions between adjoining detrusor myocytes.
    d. it has a circadian rhythm of expression.
    e. c and d are correct.
28. Which urethral mechanisms are involved in maintenance of continence in the female?
    a. Network of submucosal vascularity
    b. Guarding reflex
    c. Hammock hypothesis
    d. Sympathetic tone
    e. All are correct
29. G-coupled proteins:
    a. mediate many different receptor functions.
    b. enzymatically cleave ATP to adenosine diphosphate (ADP).
    c. regulate protein folding in the endoplasmic reticulum (ER).
    d. help to regulate intracellular calcium concentrations.

e. a and d are correct.

30. Urothelial barrier function is maintained by which of the following?
    a. Gap junctions
    b. Uroplakins
    c. TLR4 receptor
    d. Aquaporin channels
    e. Mucinous layer
31. In supraspinal spinal cord injury (SCI), mechanisms associated with neurogenic detrusor overactivity include:
    a. activity of C-fiber afferent fibers.
    b. nerve growth factor (NGF).
    c. activation of G-coupled protein receptors on detrusor muscle.
    d. contractions of the external sphincter.
    e. all are correct.
32. Membrane potential of a cell:
    a. requires energy to maintain, even at rest.
    b. at rest, is maintained by low intracellular potassium and high extracellular potassium.
    c. at rest, is maintained by low intracellular sodium and high intracellular chloride.
    d. becomes electrically more negative during an action potential in a neuron.
    e. changes during an action potential due to influx of anions into the detrusor myocyte.
33. Which of the following mechanisms is unique to M2 muscarinic receptor activation when compared to M3 muscarinic receptor activation?
    a. Mediates rise in intracellular calcium when activated
    b. Uses G-coupled proteins
    c. Involves cAMP downstream
    d. Causes detrusor contraction
    e. Causes detrusor relaxation
34. Isolated nocturia complaints are due to:
    a. nocturnal polyuria.
    b. sleep apnea.

c. detrusor overactivity.
d. peripheral edema.
e. a, b, and d are correct.

35. Which of the following declines with aging?
    a. Bladder sensation
    b. ATP content of bladder
    c. Detrusor contractile responses to α adrenergic stimulation
    d. Detrusor contractile responses to cholinergic or electrical stimulation
    e. Bladder capacity
36. Urine from bladder pain syndrome/interstitial cystitis (BPS/IC) patients has been found to contain:
    a. a protein that inhibits urothelial cell growth in culture.
    b. a virus that induces T cell–mediated inflammation.
    c. a protozoan that invades the urothelial cell.
    d. an increased level of ATP.
    e. a and d are correct.
37. The "motor-sensory" hypothesis is used to explain mechanism of action in treating:
    a. bladder underactivity with a muscarinic agonist.
    b. neurogenic detrusor overactivity with TRPV1 blocker.
    c. idiopathic detrusor overactivity with onabotulinumtoxinA.
    d. urinary urgency with antimuscarinics.
    e. urinary frequency with α-blockers.
38. The reason that some women void without measurable increase in Pdet is because:
    a. the detrusor does not need to contract during voiding in women.
    b. the urethra contains smooth, in addition to striated, muscle fibers.
    c. there is a reduced parasympathetic innervation to the female bladder.
    d. Pdet is not the only measure of the bladder's mechanical work ability.
    e. the female bladder has increased viscoelastic properties.
39. Differences between smooth and striated muscles include:

a. actinomyosin cross-bridge cycling in smooth muscle only.
b. visible striations in striated muscle only.
c. release of acetylcholine by prejunctional motor neurons in smooth muscle only.
d. lack of intermediate filaments in skeletal muscles only.
e. b and d are correct.

40. Bladder outlet obstruction results in:
    a. change in collagen subtype proportions.
    b. afferent and efferent neuronal hypertrophy.
    c. urothelial hyperplasia.
    d. C-fiber–mediated micturition reflex.
    e. a, b, and d are correct.
41. The principle behind neuromodulation in treating overactive bladder is:
    a. inhibition of detrusor interstitial cell activity.
    b. block of release of postganglionic neuronal acetylcholine.
    c. activation of C-fiber afferents.
    d. inhibition of somatic afferent processing in spinal cord.
    e. excitation of sacral sympathetic nerve fibers.
42. Which of the following animal models best mimics human BPS/IC?
    a. Social stress model
    b. Water avoidance stress model
    c. Acetic acid infusion into bladder
    d. Cyclophosphamide intraperitoneal injection
    e. Bowel radiation model
43. Etiologic causes for stress urinary incontinence may include:
    a. decreased urethral support.
    b. loss of urothelial seal.
    c. decreased serotonin in sacral spinal cord.
    d. shortened urethra.
    e. a, b, and c are correct.
44. The neurotransmitter released by sympathetic preganglionic neurons at the ganglia is:
    a. acetylcholine.
    b. norepinephrine.

c. ATP.
d. epinephrine.
e. NO.

45. The action potential in either an afferent or efferent neuron is due to:
    a. influx of Na.
    b. influx of K.
    c. influx of Cl.
    d. influx of Ca.
    e. efflux of Ca.

Answers

1. **b. Sympathetic nerves inhibit the bladder body and excite the bladder neck and urethra.**
2. **e. All are correct.**
3. **e. All of the above.**
4. **e. a, b, and c only.**
5. **a. This is an example of cross-organ sensitization.**
6. **e. All are correct.**
7. **e. All are correct.**
8. **d. b and c are true.**
9. **e. All are true.**
10. **e. a, c, and d are correct.**
11. **e. All of the above.**
12. **e. a, b, and c are correct.**
13. **a. Switching between bladder storage and emptying can occur involuntarily (reflex emptying) or voluntarily.**
14. **e. a and d are correct.**
15. **e. All of the above.**
16. **e. All of the above.**
17. **d. All of the above.**
18. **a. inositol triphosphate (ISP) hydrolysis and release of intracellular calcium.**
19. **d. All of the above.**

20. **e. a and b are correct.**
21. **e. c and d are correct.**
22. **e. c and d are correct.**
23. **a. Blockade of intracellular vesicle fusion in presynaptic nerves.**
24. **e. c and d are correct.**
25. **d. TLR4**
26. **e. a and b are correct.**
27. **e. c and d are correct.**
28. **e. All are correct.**
29. **e. a and d are correct.**
30. **b. Uroplakins.**
31. **e. All are correct.**
32. **a. Requires energy to maintain, even at rest.**
33. **c. Involves cAMP downstream.**
34. **e. a, b, and d are correct.**
35. **a. Bladder sensation.**
36. **e. a and d are correct.**
37. **d. Urinary urgency with antimuscarinics.**
38. **d. Pdet is not the only measure of bladder's mechanical work ability.**
39. **e. b and d are correct.**
40. **e. a, b, and d are correct.**
41. **d. Inhibition of somatic afferent processing in spinal cord.**
42. **b. Water avoidance stress model.**
43. **e. a, b, and c are correct.**
44. **a. Acetylcholine.**
45. **a. Influx of Na.**

Chapter Review

1. The bladder has two parts: the body, which lies above the ureteral orifices, and the base, consisting of the trigone and bladder neck.
2. Smooth muscle is able to adjust its length over a much wider range than skeletal muscle. Thus an empty bladder has a small intravesical space despite the amount of smooth muscle it contains.

3. There is a complete, competent ring of smooth muscle around the bladder neck in the male. This does not occur in the female.
4. In women the density of adrenergic innervation in the bladder neck is less than in men.
5. Myofibroblasts in the LP modulate physiologic interactions between the urothelium and detrusor.
6. Bladder wall blood flow is reduced by distention; in patients with decreased compliance, this effect is pronounced.
7. In the female, the external urethral sphincter covers the ventral surface of the urethra in a horseshoe configuration.
8. The levator ani pelvic floor muscle does not surround the ventral aspect of the urethra in either the male or the female.
9. The external urethral sphincter is composed of (1) periurethral striated muscle of the pelvic floor and (2) striated muscle within the urethra.
10. The bladder urothelium serves a barrier function but is permeable to water to a limited degree and can actively transport sodium.
11. There is no definite evidence that the glycosaminoglycan (GAG) layer acts as a primary epithelial barrier.
12. Urothelial cells release chemical mediators such as NO, ATP, acetylcholine, and substance P that have excitatory and inhibitory actions on afferent nerves in the bladder wall.
13. Prostaglandins are released from the urothelium.
14. Uroplakin and tight junction proteins are important in maintaining the urothelial barrier function.
15. The normal bladder at rest may be spontaneously active.
16. A low voiding pressure in women does not equate with impaired detrusor contractility.
17. The parasympathetic nerves from S2 to S4 excite the bladder and relax the urethra. They have afferent fibers. The lumbar sympathetic nerves inhibit the bladder body and excite the bladder base and urethra. They also have afferent fibers. The pudendal nerves (S2 to S4) excite the external urethral sphincter. Afferent nerve fibers travel with the pudendal nerve as well.
18. There may be parasympathetic afferent and efferent nerve interconnections at the level of the intramural ganglia.

19. Pelvic nerve afferents monitor bladder volume and amplitude of the bladder contraction.
20. The bladder neck and proximal urethra contain the largest density of bladder nerves.
21. Decreased afferent sensitivity or excitability in certain pathologic conditions, as well as aging, may be an important cause of impaired voiding.
22. Activation of the parasympathetic pathway during voiding triggers the release of NO, which is a major inhibitor of urethral smooth muscle.
23. Cross organization may occur between bladder and bowel, uterus, pelvic urethra, vagina, and prostate. This may contribute to the chronic pelvic pain syndrome.
24. A substantial proportion of the C-fiber afferent population is silent; pathologic conditions may recruit mechanosensitive C-fibers to form a new functional afferent pathway.
25. Activation of the central serotonergic system can suppress voiding by inhibiting the parasympathetic excitatory input to the urinary bladder.
26. The bladder sympathetic reflex promotes closure of the urethral outlet and inhibits neurally mediated contractions of the bladder.
27. While the bladder fills, the external sphincter activity increases (guarding reflex)—i.e., pudendal motor neurons are activated by bladder afferent input.
28. The dorsal pontine tegmentum is the control center for micturition.
29. Stimulation of β-2 and β-3 receptors relaxes the detrusor.
30. Sex steroids modulate receptors and influence growth of bladder tissues.
31. After SCI, a C-fiber–mediated spinal reflex develops and may play a role in the development of detrusor overactivity.
32. Alterations of neural networks occur in the central nervous system following obstruction of the lower urinary tract.
33. Obstruction-induced detrusor overactivity may be due to denervation supersensitivity.
34. With aging detrusor contractility, bladder sensation and urethral pressure decline.

35. Sacral neuromodulation is thought to have its beneficial effect by somatic inhibition of sensory processing in the spinal cord.
36. The hammock hypothesis of urinary continence suggests that the urethra has a fixed dorsal surface due to its attachments to the pubis, pelvic muscles, and fascia, which allow ventral wall compression of the urethra against the fixed dorsal wall.